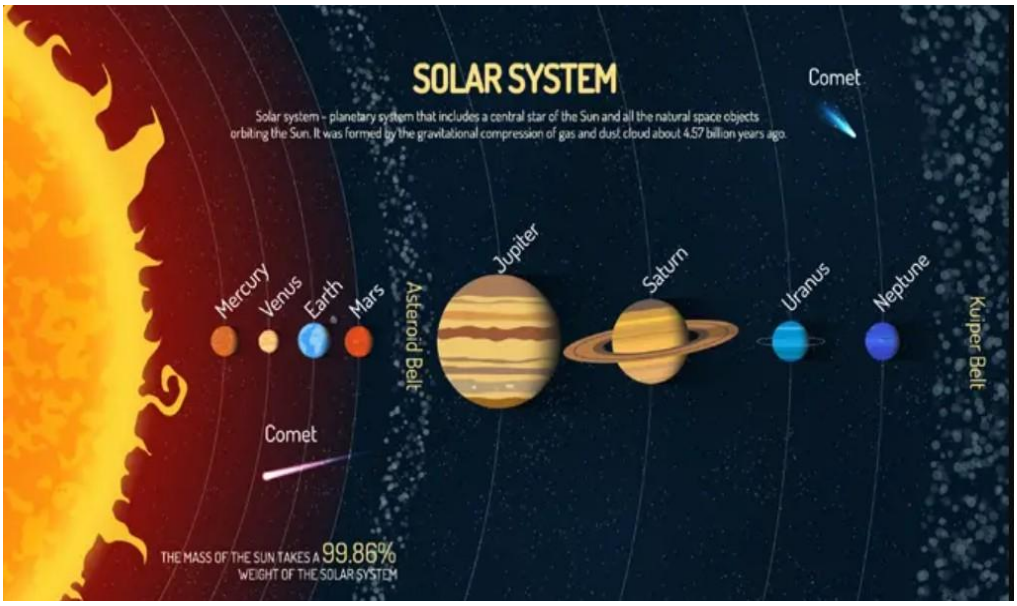
8 THE SOLAR SYSTEM



LEARNING OUTCOMES

The learner should be able to:

- know the relative sizes, positions, and motions of the earth, sun and moon.
- understand how day and night occur and demonstrate the phases of the moon.
- understand the roles of the sun, earth and moon in explaining time, seasons, eclipses, and ocean tides.
- know the components of the solar system and their positions.
- know the main characteristics of the inner and outer planets in the solar system.
- understand the various views about the origin and structure of the universe.

INTRODUCTION

The Universe.

The Universe is all of space. It contains many galaxies separated by vast expanses of empty space.

A galaxy is a large cluster of thousands, or millions of stars (e.g. our galaxy is the Milky Way).

A star is a large body of matter that is undergoing nuclear fusion and emitting light and heat.
The Sun is a star and its one of the very many stars in our milk way galaxy.

The solar system is a collection of celestial bodies, including the sun, planets, moons, asteroids, and comets, bound together by gravity.

The sun and many other stars have a solar system. A solar system consists of a central star and all the objects held to the star by gravity.

There are eight recognized planets in our solar system, in order of their distance from the sun, are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. Earth is the third planet from the sun and is the only known celestial body to support life.

A planet is a large body in orbit around a star. A moon is a natural satellite of a planet. Earth has one moon.
The planets are divided into two groups

The inner planets (terrestrial planets).
They are smaller, closer to the sun, and have rocky surfaces (Mercury, Venus, Earth, mars)

The outer planets (jovian planets).
They are larger, farther from the sun and do not have solid surfaces (Jupiter, Saturn, Uranus, Neptune)

Days and nights are brought about by the rotation of the earth about its axis. the day time is not always always equal to night time except during equinox.

The **tilting of earth's surface** of rotation causes variation in the distance of a place from the sun. This variation in turn leads to changes in The amount of sunlight received at different points of the earth. This has an implication on the characteristics and weather in these places and have lead to seasons like summer, autumn, winter, spring.

ACTIVITIES

1. A student is curious about why we experience day and night on Earth. Explain the physics behind the rotation of the Earth on its axis and how it creates the cycle of day and night.

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2. An astronaut is preparing for a mission to the International Space Station (ISS). Discuss how Earth's magnetic field provides protection to astronauts from harmful solar and cosmic radiation while in space.

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3. A space agency is planning a mission to land on Mars or jupiter. Discuss the physics challenges and considerations

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4. You are in a classroom and you want to show how the Earth is a magnet and how a compass works. You have a globe, a bar magnet, a compass, and some iron filings.
   How can you use the globe and the bar magnet to model the Earth's magnetic field?
   How can you use the compass and the iron filings to visualize the magnetic field lines?

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5. You are a teacher at a primary school, and you want to demonstrate the phases of the moon to your students using a lamp, a ball, and a stick.
   How would you set up the experiment?
   What would you ask your students to observe and record?
   How would you explain the causes of the moon phases?

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6. You are a philosopher, and you are interested in the origin and structure of the universe.
   What are the main views or theories that have been proposed to explain the origin and structure of the universe?
   How do they differ in their assumptions, methods, and evidence?

What are the strengths and weaknesses of each view or theory? How do they relate to your own beliefs or values?

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7. You are a futurist, and you are interested in the possibilities and challenges of exploring and colonizing other planets or moons in the solar system. You want to write a report or a presentation that outlines the current and future plans and projects of various space agencies and organizations, such as NASA, ESA, or SpaceX.
What are the main goals and motivations of these plans and projects?
What are the main technical and ethical issues that they face?

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8. on a particular day in your village, it was about 3:00 pm. The whole village turned dark without any sunshine. Members of the community were very scared, some of them thought that the day of judgement had come and others thought that the village had been bewitched. As a person who has knowledge about the solar system.

Task
write down a report, explaining to the citizens what had caused the darkness.

9. On November 7’ 2022, Uganda launched its first satellite named Pear/AfricaSat-1 into space with the help of National Aeronautics and Space Administration (NASA). The purpose of this mission was to study weather patterns. Students of Physics were availed with data collected over a certain period of time and they noticed the following. While some places were having day time, other places were having night time. Various places were having different seasons.

Task
As a learner of Physics explain;

why some places had daytime while it was night time at other places, why different places had different weather patterns.

10. The world population is growing so fast that different governments have expressed concern that in the near future the available land on Earth may not be sufficient for the economic activities required to sustain the growing population. There is need to explore the planets in our solar system with the view of identifying another planet hat can sustain life.

Support

Imagine that you are a scientist or an engineer designing a new pace probe to explore our solar system for this purpose.
Watch a video on ”space probes”.

Task

Make a plan and create your own design of a space probe that can enable you to review the weather conditions on the different planet of our solar system.

11. While reading a Science magazine, Atwine came across the following paragraph:

    ”The Earth is part of the solar system in the universe. It spins about a tilted axis as it revolves around the Sun, leading to the manifestation of all kinds of phenomena, such as day and night time, seasons, and those with rare names such as aphelion and apogee. Oh, and not forgetting, the solstice and the tides that rise and fall according to the phases of the Moon. Indeed, our planet is as strange as the theories that try to explain how it came into existence!”

    Atwine was utterly confused by this paragraph.

    Support

    

    Task
    Supposing she brought it to you for explanation, prepare a written response for him.